1.1. Escape velocity ∗

As given below in Exercise 1.9, show that the escape velocity from the earth is $v = \sqrt{2GM_{\mathrm{E}}/R}$, up to numerical factors. You can use the fact that the form of Newton's gravitation force law implies that the acceleration (and hence overall motion) of the particle doesn't depend on its mass.

1.2. Mass in a tube *

A tube of mass M and length ℓ is free to swing around a pivot at one end. A mass m is positioned inside the (frictionless) tube at this end. The tube is held horizontal and then released (see Fig. 1.5). Let η be the fraction of the tube that the mass has traversed by the time the tube becomes vertical. Does η depend on ℓ?

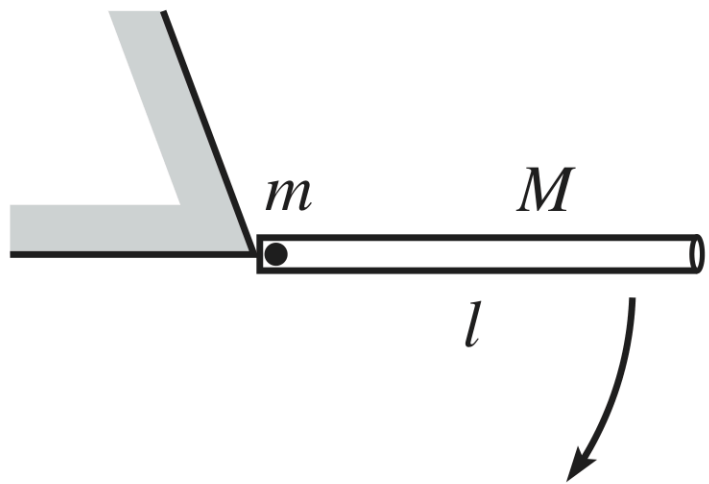


Fig. 1.5

1.3. **Waves in a fluid** *

How does the speed of waves in a fluid depend on its density, ρ, and "bulk modulus," B (which has units of pressure, which is force per area)?

1.4. **Vibrating star** *

Consider a vibrating star, whose frequency ν depends (at most) on its radius R, mass density ρ, and Newton's gravitational constant G. How does ν depend on R, ρ, and G?

1.5. **Damping** **

A particle with mass m and initial speed V is subject to a velocity-dependent damping force of the form bv^n.

(a) For $n = 0, 1, 2, \ldots$, determine how the stopping time depends on m, V, and b.

(b) For $n = 0, 1, 2, \ldots$, determine how the stopping distance depends on m, V, and b.

Be careful! See if your answers make sense. Dimensional analysis gives the answer only up to a numerical factor. This is a tricky problem, so don't let it discourage you from using dimensional analysis. Most applications of dimensional analysis are quite straightforward.

1.6. **Projectile distance** *

A person throws a ball (at an angle of her choosing, to achieve the maximum distance) with speed v from the edge of a cliff of height h. Assuming that one of the following quantities is the maximum horizontal distance the ball can travel, which one is it? (Don't solve the problem from scratch, just check special cases.)

$$\frac{gh^2}{v^2}, \quad \frac{v^2}{g}, \quad \sqrt{\frac{v^2 h}{g}}, \quad \frac{v^2}{g}\sqrt{1+\frac{2gh}{v^2}}, \quad \frac{v^2}{g}\left(1+\frac{2gh}{v^2}\right), \quad \frac{v^2/g}{1-\frac{2gh}{v^2}}.$$

1.7. Two masses, one swinging **

Two equal masses are connected by a string that hangs over two pulleys (of negligible size), as shown in Fig. 1.6. The left mass moves in a vertical line, but the right mass is free to swing back and forth in the plane of the masses and pulleys. It can be shown (see Problem 6.4) that the equations of motion for r and θ (labeled in the figure) are

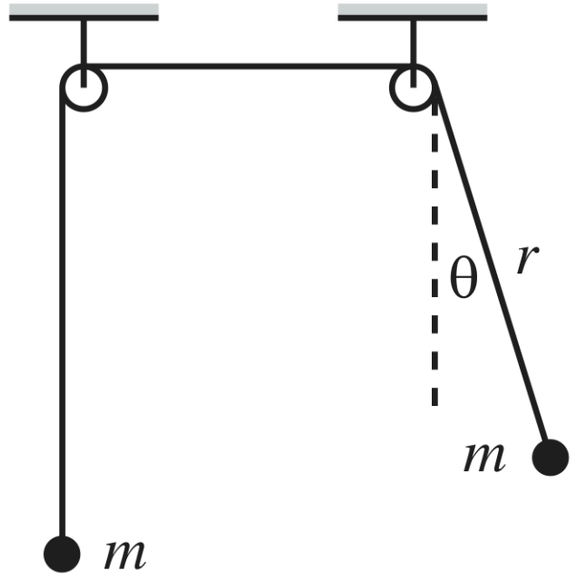


Fig. 1.6

$$\begin{aligned} 2\ddot{r} &= r\dot{\theta}^2 - g(1-\cos\theta), \\ \ddot{\theta} &= -\frac{2\dot{r}\dot{\theta}}{r} - \frac{g\sin\theta}{r}. \end{aligned} \tag{1.16}$$

Assume that both masses start out at rest, with the right mass making an initial angle of $10^\circ = \pi/18$ with the vertical. If the initial value of r is 1 m, how much time does it take for it to reach a length of 2 m? Write a program to solve this numerically. Use $g = 9.8\,\text{m/s}^2$.

1.8. **Pendulum on the moon**

If a pendulum has a period of 3 s on the earth, what would its period be if it were placed on the moon? Use $g_M/g_E \approx 1/6$.

1.9. **Escape velocity** *

The *escape velocity* on the surface of a planet is given by

$$v = \sqrt{\frac{2GM}{R}}, \tag{1.17}$$

where M and R are the mass and radius of the planet, respectively, and G is Newton's gravitational constant. (The escape velocity is the velocity needed to refute the "What goes up must come down" maxim, neglecting air resistance.)

(a) Write v in terms of the average mass density ρ, instead of M.

(b) Assuming that the average density of the earth is four times that of Jupiter, and that the radius of Jupiter is 11 times that of the earth, what is $v_{\mathrm{J}}/v_{\mathrm{E}}$?

1.10. **Downhill projectile** *

A hill is sloped downward at an angle θ with respect to the horizontal. A projectile of mass m is fired with speed v_0 perpendicular to the hill. When it eventually lands on the hill, let its velocity make an angle β with respect to the horizontal. Which of the quantities θ, m, v_0, and g does the angle β depend on?

1.11. Waves on a string *

How does the speed of waves on a string depend on its mass M, length L, and tension (that is, force) T?

1.12. **Vibrating water drop** *

Consider a vibrating water drop, whose frequency ν depends on its radius R, mass density ρ, and surface tension S. The units of surface tension are (force)/(length). How does ν depend on R, ρ, and S?

1.13. **Atwood's machine** *

Consider the "Atwood's" machine shown in Fig. 1.7, consisting of three masses and three frictionless pulleys. It can be shown that the acceleration of m_1 is given by (just accept this):

$$a_1 = g\frac{3m_2m_3 - m_1(4m_3 + m_2)}{m_2m_3 + m_1(4m_3 + m_2)}, \tag{1.18}$$

with upward taken to be positive. Find a_1 in the following special cases:

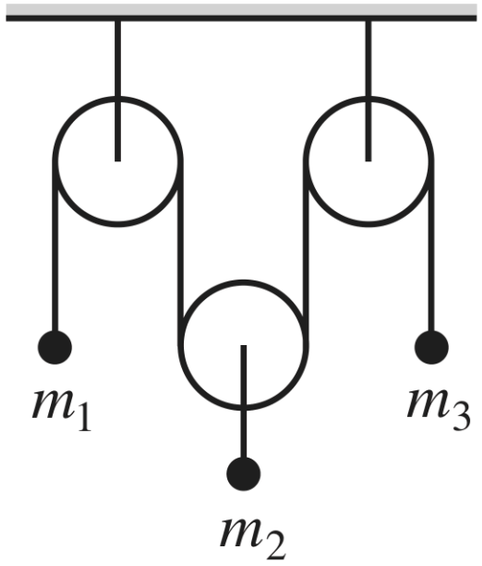


Fig. 1.7

(a) $m_2 = 2m_1 = 2m_3$.
(b) m_1 much larger than both m_2 and m_3.
(c) m_1 much smaller than both m_2 and m_3.
(d) $m_2 \gg m_1 = m_3$.
(e) $m_1 = m_2 = m_3$.

1.14. Cone frustum *

A cone frustum has base radius b, top radius a, and height h, as shown in Fig. 1.8. Assuming that one of the following quantities is the volume of the frustum, which one is it? (Don't solve the problem from scratch, just check special cases.)

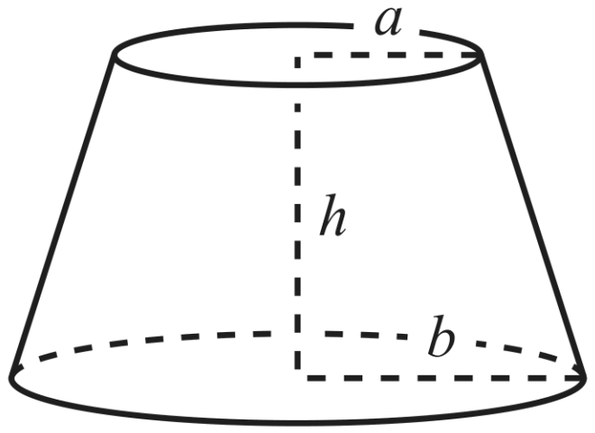


Fig. 1.8

$$\frac{\pi h}{3}(a^2+b^2), \quad \frac{\pi h}{2}(a^2+b^2), \quad \frac{\pi h}{3}(a^2+ab+b^2),$$

$$\frac{\pi h}{3}\cdot\frac{a^4+b^4}{a^2+b^2}, \quad \pi hab.$$

1.15. Landing at the corner *

A ball is thrown at an angle θ up to the top of a cliff of height L, from a point a distance L from the base, as shown in Fig. 1.9. Assuming that one of the following quantities is the initial speed required to make the ball hit right at the edge of the cliff, which one is it? (Don't solve the problem from scratch, just check special cases.)

$$\sqrt{\frac{gL}{2(\tan\theta - 1)}}, \quad \frac{1}{\cos\theta}\sqrt{\frac{gL}{2(\tan\theta - 1)}}, \quad \frac{1}{\cos\theta}\sqrt{\frac{gL}{2(\tan\theta + 1)}},$$

$$\sqrt{\frac{gL\tan\theta}{2(\tan\theta + 1)}}.$$

Fig. 1.8

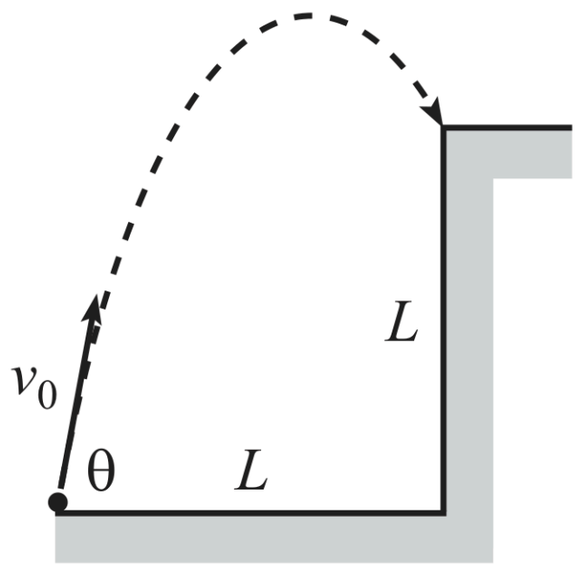


Fig. 1.9

1.16. Projectile with drag **

Consider a projectile subject to a drag force $\mathbf{F} = -m\alpha\mathbf{v}$. If it is fired with speed v_0 at an angle θ, it can be shown that the height as a function of time is given by (just accept this here; it's one of the tasks of Exercise 3.53)

$$y(t) = \frac{1}{\alpha}\left(v_0 \sin\theta + \frac{g}{\alpha}\right)\left(1 - e^{-\alpha t}\right) - \frac{gt}{\alpha}. \tag{1.19}$$

Show that this reduces to the usual projectile expression, $y(t) = (v_0 \sin\theta)t - gt^2/2$, in the limit of small α. What exactly is meant by "small α"?

1.17. Pendulum **

A pendulum of length ℓ is released from the horizontal position. It can be shown that the tangential $F = ma$ equation is (where θ is measured with respect to the vertical)

$$\ddot{\theta} = -\frac{g \sin\theta}{\ell}. \tag{1.20}$$

If $\ell = 1\,\text{m}$, and $g = 9.8\,\text{m/s}^2$, write a program to show that the time it takes the pendulum to swing down through the vertical position is $t \approx 0.592\,\text{s}$. This happens to be about 1.18 times the $(\pi/2)\sqrt{\ell/g} \approx 0.502\,\text{s}$ it would take the pendulum to swing down if it were released from very close to the vertical (this is 1/4 of the standard period of $2\pi\sqrt{\ell/g}$ for a pendulum). It also happens to be about 1.31 times the $\sqrt{2\ell/g} \approx 0.452\,\text{s}$ it would take a mass to simply freefall a height ℓ.

1.18. Distance with damping **

A mass is subject to a damping force proportional to its velocity, which means that the equation of motion takes the form $\ddot{x} = -A\dot{x}$, where A is some constant. If the initial speed is 2 m/s, and if $A = 1\ \text{s}^{-1}$, how far has the mass traveled at 1 s? 10 s? 100 s? You should find that the distance approaches a limiting value.

Now assume that the mass is subject to a damping force proportional to the square of its velocity, which means that the equation of motion now takes the form $\ddot{x} = -A\dot{x}^2$, where A is some constant. If the initial speed is 2 m/s, and if $A = 1\ \text{m}^{-1}$, how far has the mass traveled at 1 s? 10 s? 100 s? How about some larger powers of 10? You should find that the distance keeps growing, but slowly like the log of t. (The results for these two forms of the damping are consistent with the results of Problem 1.5.)